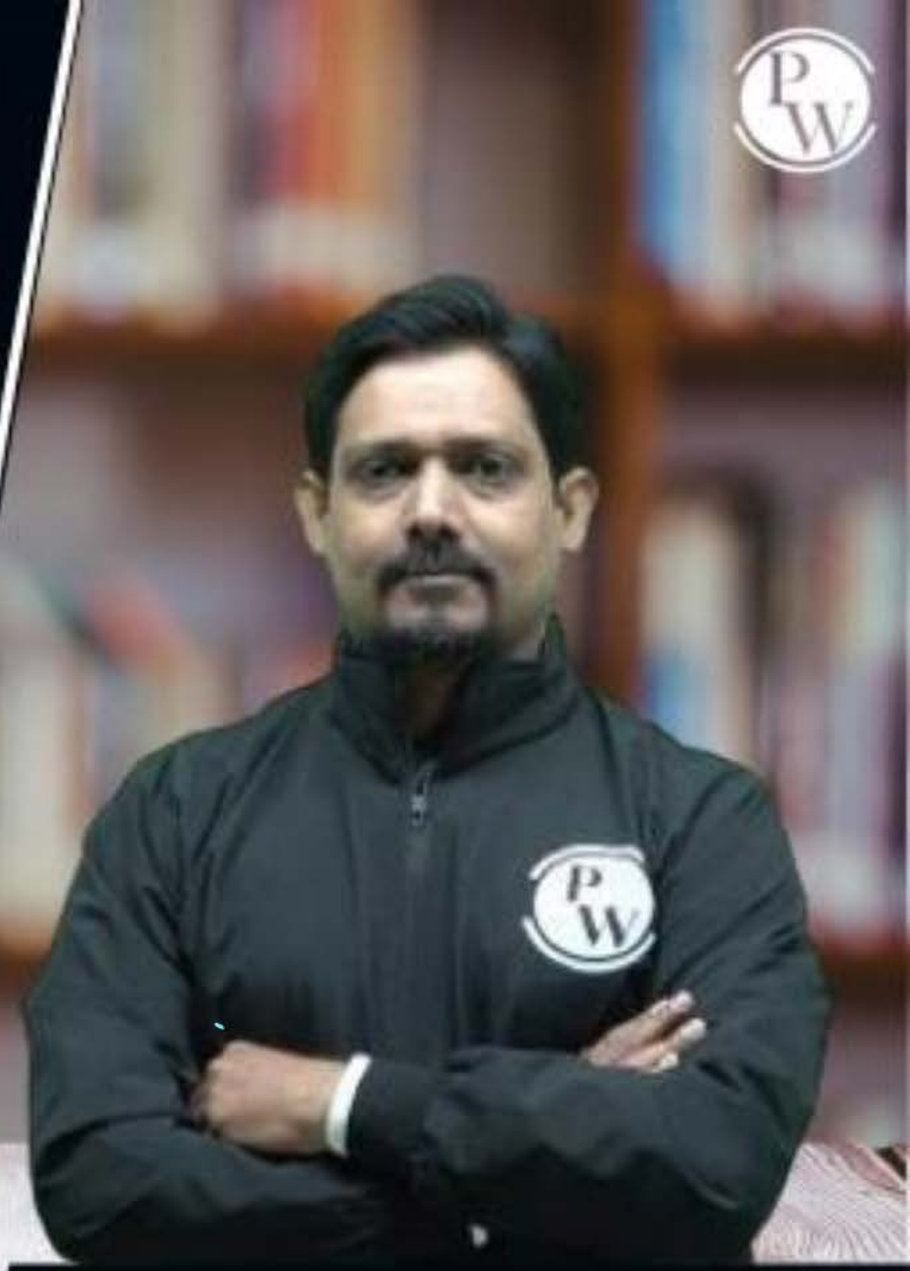
Computer Science & IT



Data Structure & Programming

Tree

Lecture No. 02



By- Abhishek Sir

Recap of Previous Lecture



Topic	Tree
Topic	general Tree, forest
Topic	k-ary Tree
Topic	Theorem
Topic	

Slide

Topics to be Covered



Topic	Theorem
Topic	Binary Tree
Topic	Special Tree
Topic	Height of Binary Tree
Topic	Counting of unlabelled Tree

Slide



Topic : Tree

K ary tree
Binary tree
Height of binary tree
Binary tree theorem
Special tree
Counting unlabeled node



Topic : Tree

No. of edges a Node connected-is degree



3-ary Tree

0 or 3 children

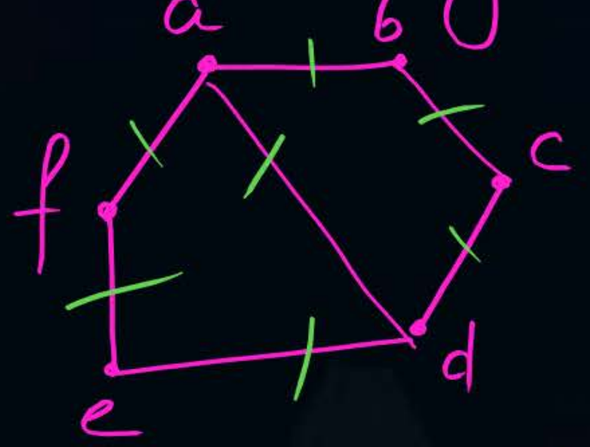
Sum of degree

$= 2|E|$

$= 3+2+3+2+2+2$

$= 14 = 2|E| \Rightarrow E = 7$

Handshaking algorithm



degree(a) = 3 degree(b) = 2

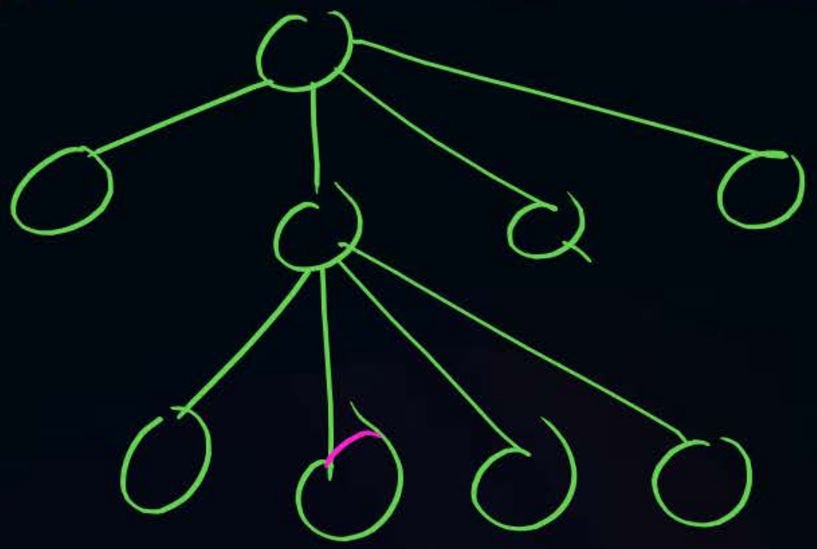
degree(c) = 2, degree(d) = 3

degree(e) = degree(f) = 2

Slide



Topic : Tree



4-ary

k-ary

L leaf node then degree of L is 1

I is Internal the degree of I is = k+1
(except Root)

Total No. of Nodes = (I + L) (n)

Total No. of edges = I + L - 1 (n-1)

Slide



Topic : Tree

Sum of degree = $2|E|$

$$L \times 1 + I(k+1) - 1 = 2(I + L - 1)$$

$$\Rightarrow \quad L + Ik + I - 1 = 2I + 2L - 2$$

$$\Rightarrow \quad 2L - L = Ik + I - 1 - 2I + 2$$

$$\Rightarrow \quad L = Ik - I + 1$$

$$\therefore \quad L = I(k-1) + 1$$

Slide



Topic : Question

0, oo

#Q A complete n-ary tree is a tree in which each node has n children. Let I be the number of internal nodes and L be the number of leaves in a complete n-ary tree. If L = 41, and I = 10, what is the value of n?

(A) 3

(B) 4

(C) 5

(D) 6

$L = I(n-1)+1$

$41 = 10(n-1)+1$

$\Rightarrow 10(n-1) = 40$

$n-1 = 4$

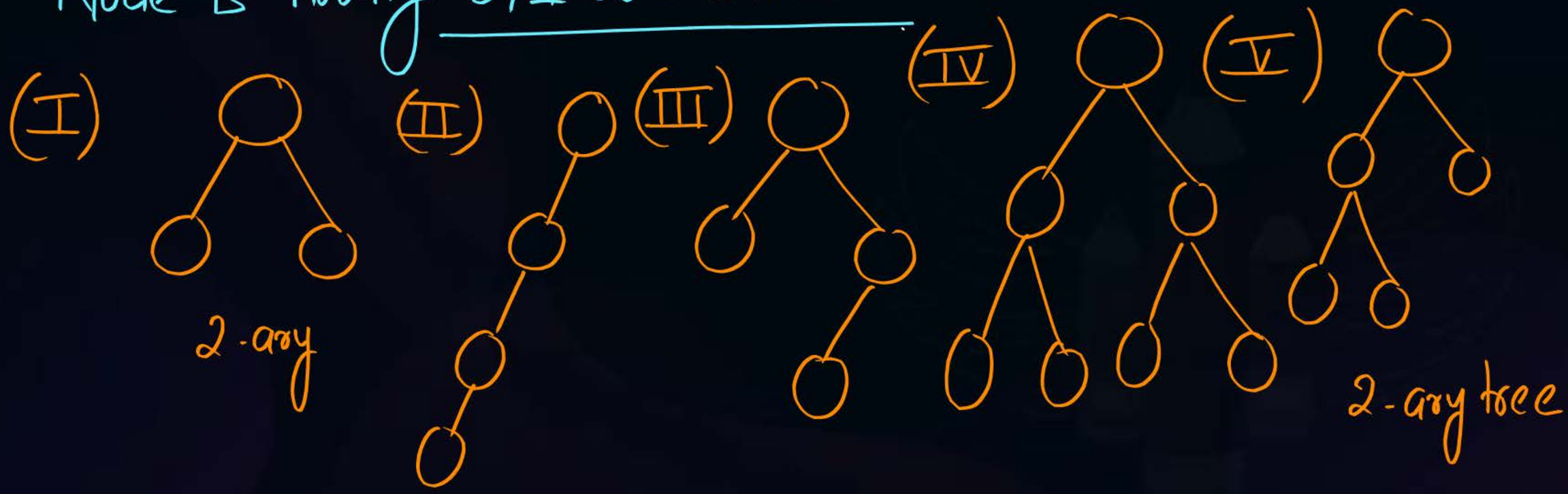
$n = 5$

Slide



Topic : Tree

Binary Tree: Binary tree is a tree in which each Node is having 0, 1 or 2 children.

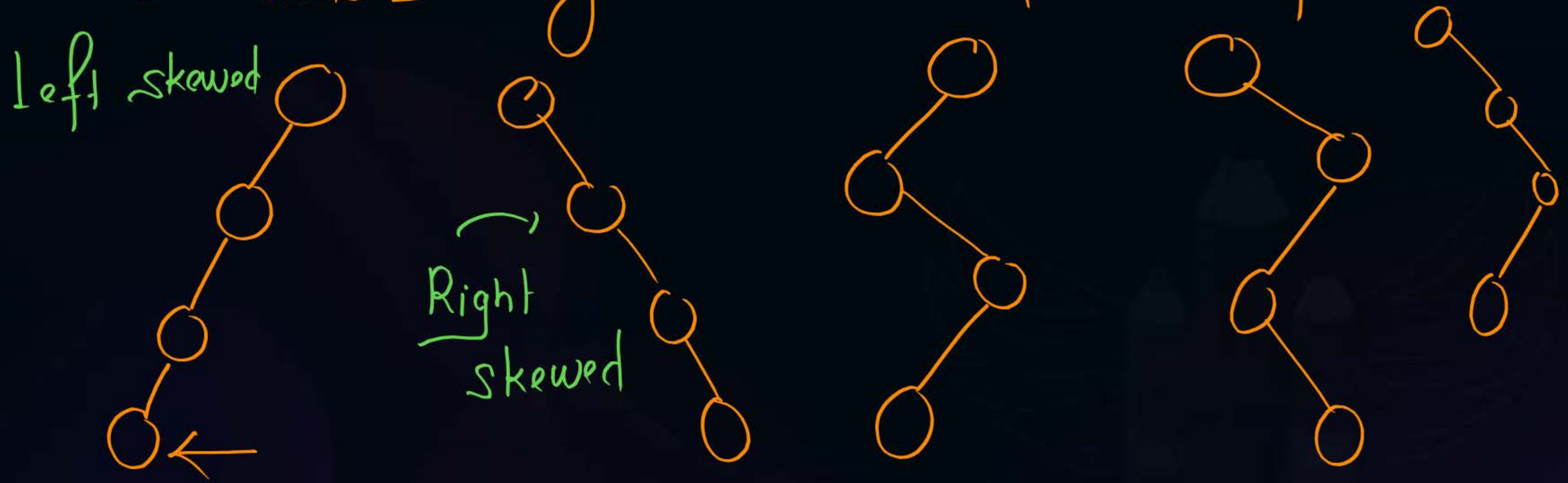


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Topic : Tree

Skewed binary tree : Skewed binary Tree is a tree in which each node is having 1 children except 1. (leaf Node)

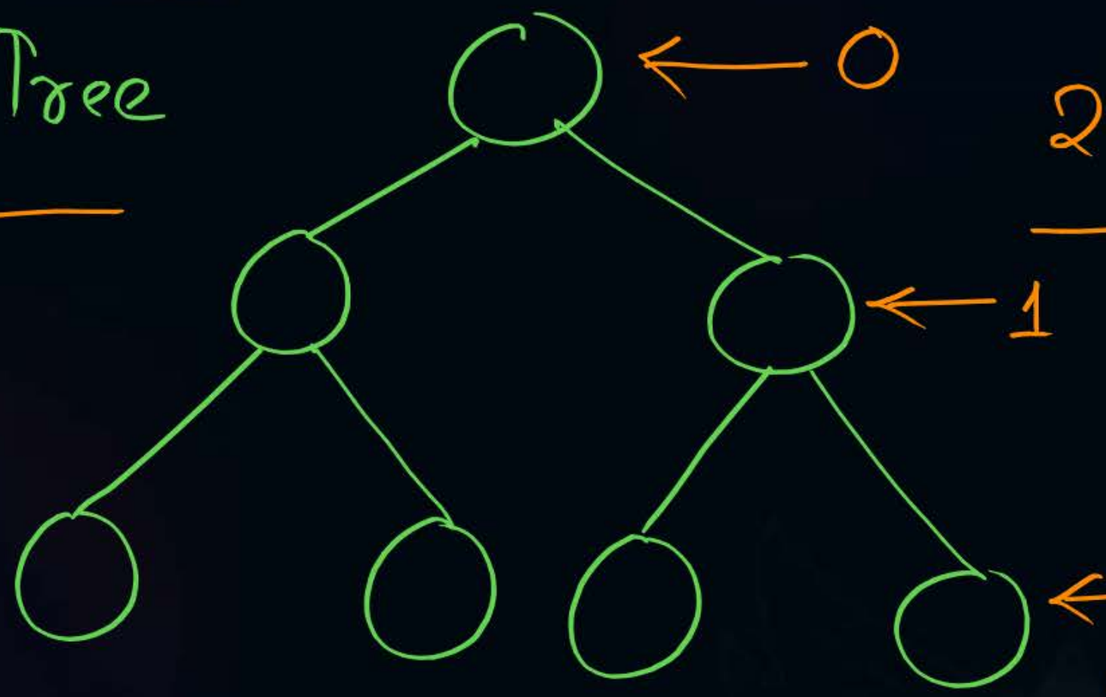


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Topic : Tree

full Binay Tree



2-ary Tree

Each Node is having 0 or 2 children

Extra: all leaves are at same height

Condition

Slide



Topic : Tree

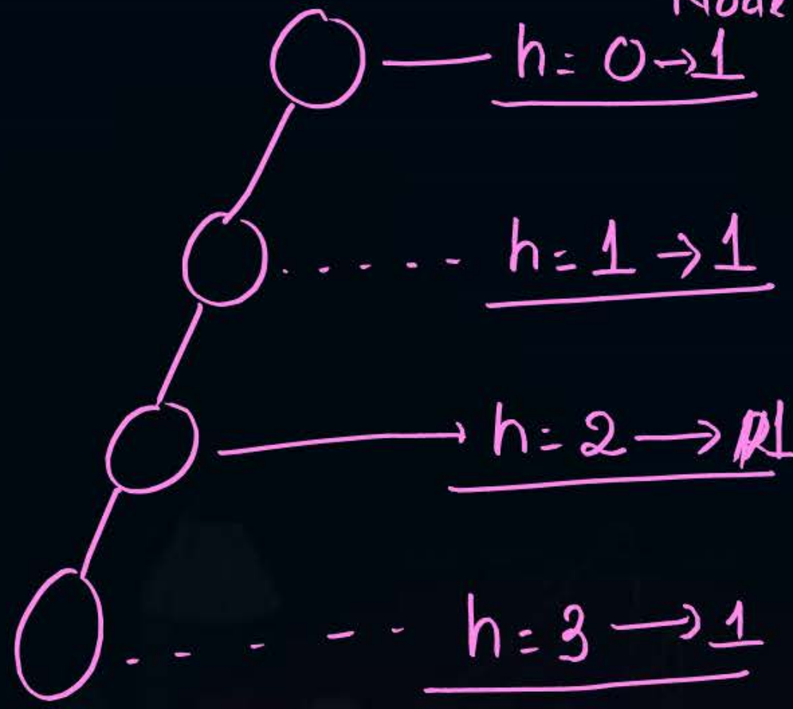
Height of Binary with n node

Maximum Height : skewed binary Tree

$$\underbrace{(1+1+1+1\ldots1)}_{(h+1)} = n$$

$$h+1 = n$$

$$\boxed{h = n-1}$$



Slide



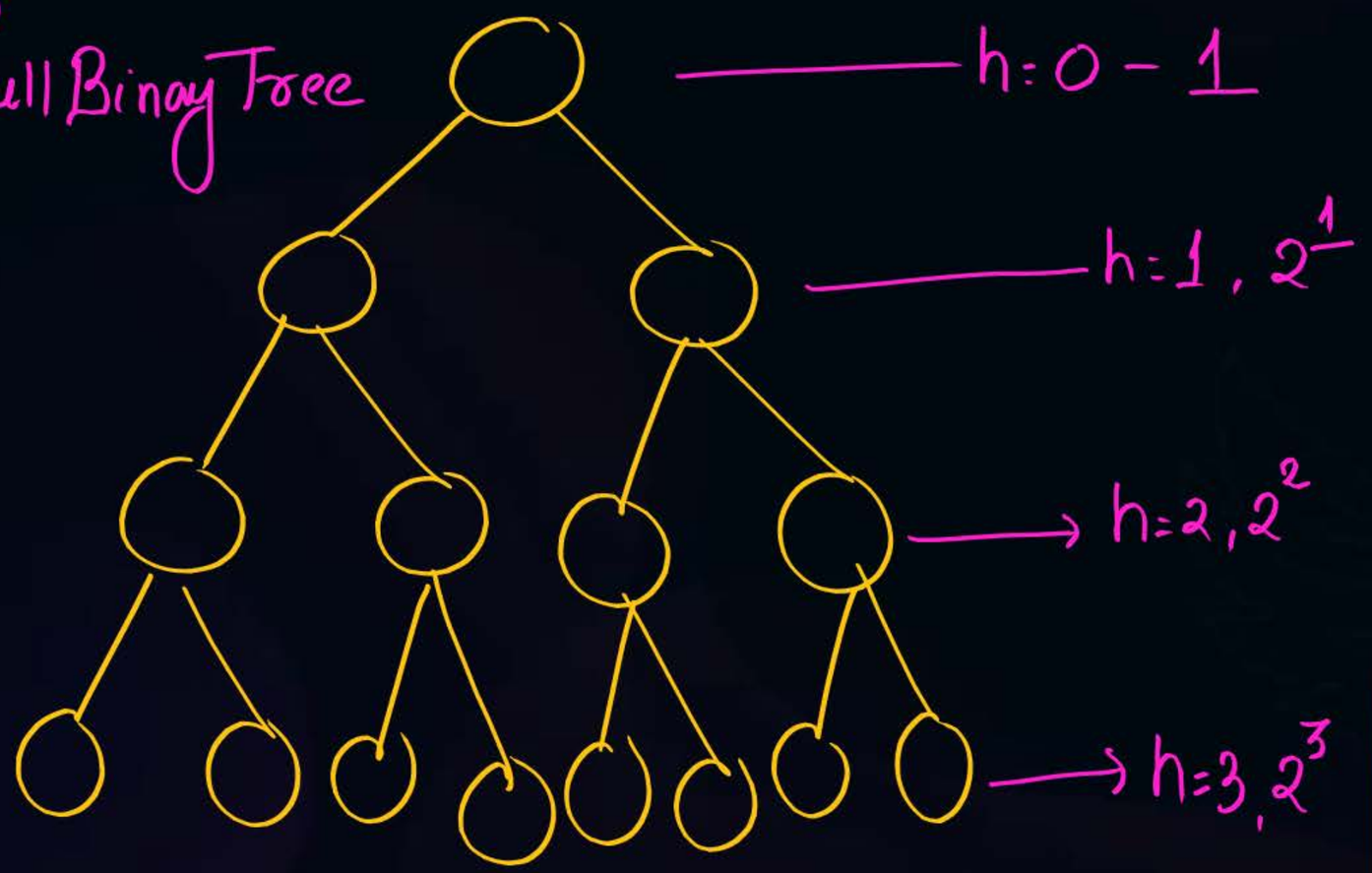
Topic : Tree

$L = I(2-1)+1$

$L = I+1$

Minimum Height with n node

full Binay Tree



$2^0+2^1+2^2+\cdots+2^h = n$

=

Slide



Topic : Tree

$\frac{a(r^n-1)}{r-1}$

$2^0+2^1+2^2-1\cdots+2^h=n$

$=\frac{1(2^{h+1}-1)}{2-1}=n$

$=2^{h+1}-1=n$

$=2^{h+1}=n+1$

$=\boxed{2^h=\frac{n+1}{2}}$

$\log_2$ both side

$\log_2 2^h=\log_2\left(\frac{n+1}{2}\right)$

$\Rightarrow h\log_2 2=\log_2\left(\frac{n+1}{2}\right)$

$\Rightarrow h=\log_2\left(\frac{n+1}{2}\right)$

Slide



Topic : Tree

Full Binary Tree

$$h = \log_2\left(\frac{n+1}{2}\right) = \log_2(n+1) - \log_2 2$$

$$= \log_2(n+1) - 1$$

$$2^h\left(\text{No. of leaves}\right) = \frac{n+1}{2} = \frac{15+1}{2} = 8$$

$$n = 15$$

Slide



Topic : Question

$x = \lfloor \log_2(127+1) \rfloor - 1$

#Q Let T be a binary tree with 127 nodes. The minimum and maximum possible heights of T are:

$\lfloor \log_2 128 \rfloor - 1$

$7 - 1 = 6$

$\lfloor \log_2 127 \rfloor = 6$

if x is minimum Height with 127 nodes

Node2 y is Maximum Height with 127 nodes

then x+y is ——

$X = 6$

Maximum : $h = n-1$

$h = 127-1 = 126$

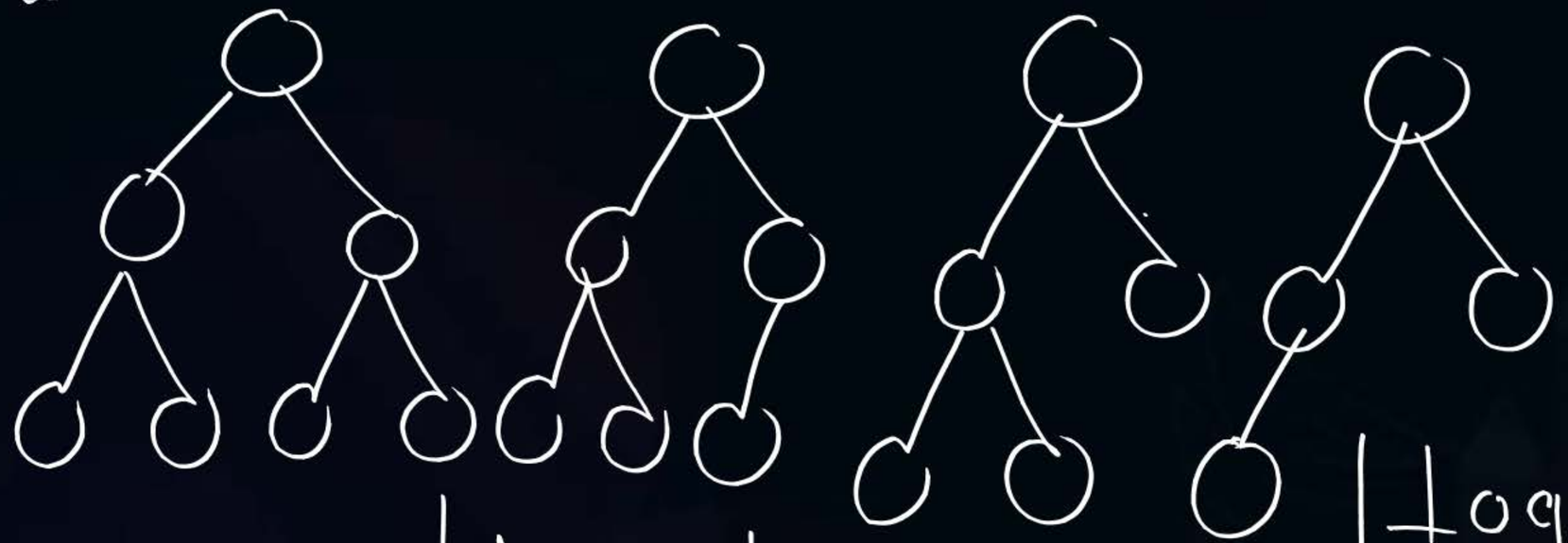
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Topic : Tree

Minimum Height of Binary Tree with n node $= \lfloor \log_2 n \rfloor$

h:-



$\lfloor \log_2 7 \rfloor = 2$

$\lfloor \log_2 6 \rfloor = 2$

$\lfloor \log_2 5 \rfloor = 2$

$\lfloor \log_2 4 \rfloor = 2$

Slide



Topic : Tree

Minimum Height of binary Tree with n node

$$\lfloor \log_2 n \rfloor$$

Slide



Topic : Tree

Counting unlabelled Binary Tree $(T(n))$ t.me/Abhishekshaarma pw

1.  $n = 1$ $T(1) = 1$

2.  $n = 2$ $T(2) = 2$

Slide



Topic : Tree

3. $n = 3$



2 Node



$T(3) = 5$

L	R
2	0
1	1
0	2





Slide



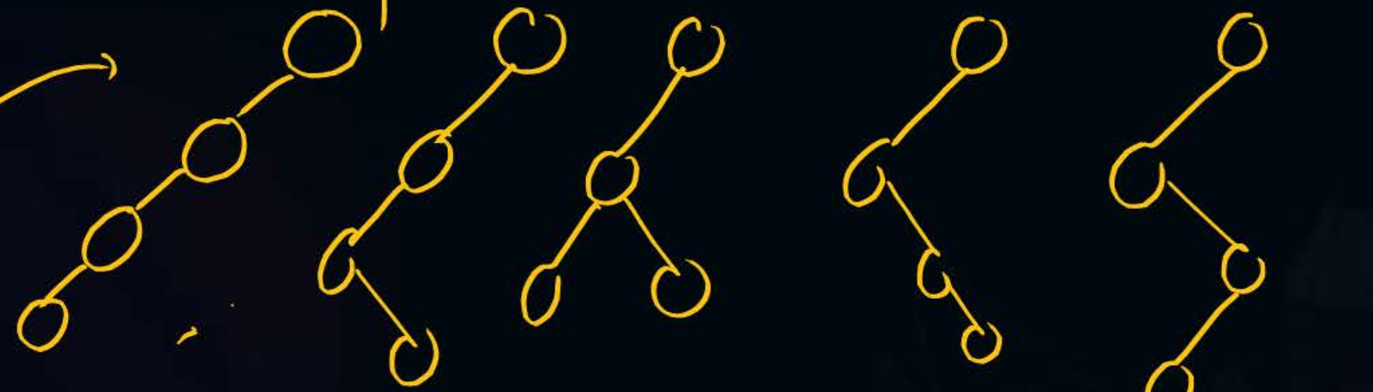
Topic : Tree

$n = 4$

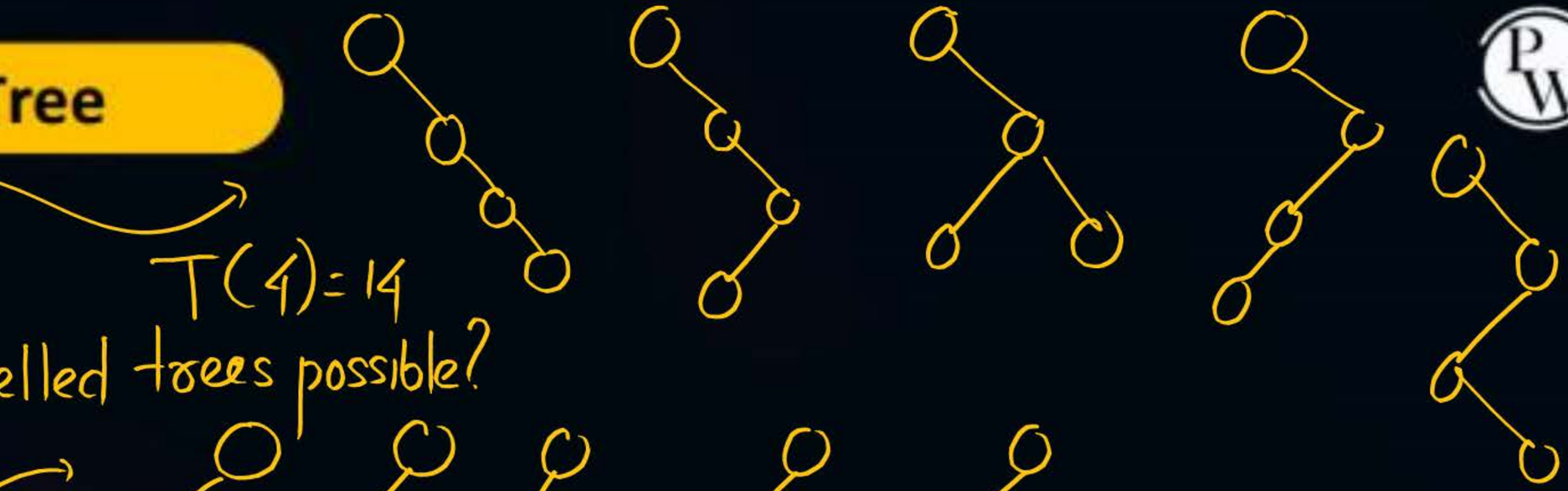
$T(4) = 14$

No. of unlabelled trees possible?

L	R
3	0
2	1
1	2
0	3







Slide



Topic : Tree



n = 4

L	R	
3	1	— 5
2	1	— 2
1	2	— 2
0	3	— 5

14

Slide



Topic : Tree

$T(1)=1 \quad T(2)=2 \quad T(3)=5 \quad T(4)=14$

$T(5)=?$ one Root node

L	R
4	0 ⟶ 14
3	1 — 5
2	2 — 2*2 : 4
1	3 5
0	4 14

$= 42$

Slide



Topic : Tree

No. of unlabelled tree with n Node

$$= \frac{1}{n+1} {}^{2n}C_n$$

Catalan No.

Slide



Topic : Question

#Q A ~~complete~~ k-ary tree is a tree in which each node has k children or no children. Let there be 24 internal node and 97 be the number of leaves. The value of the k is

$$L = I(k-1) + 1$$

$$97 = 24(k-1) + 1$$

$$24(k-1) = 97 - 1$$

$$24(k-1) = 96 \Rightarrow k-1 = 4 \Rightarrow \boxed{k=5}$$

Slide



Topic : Question

3-ary tree 0 or 3 Node

#Q Let T be complete ternary tree of height 6 where each node is having 0 or 3 children. The maximum and minimum number of nodes possible in T is (The root node is at height 0)?

(A) 13 and 3280

(B) 3280 and 13

(C) 1093 and 19

(D) 127 and 7

Slide



2 mins Summary

Topic — k-ary Tree

Topic — Binary Tree height

Topic — Counting unlabelled Trees

Topic

Topic

Slide

THANK - YOU